

Assessing Learner Readiness and Expectations for Metaverse-Based Education: A Pre-Survey Analysis of EON XR in Healthcare Service Quality Course

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Abstract: Immersive technologies, specifically augmented reality, virtual reality and Metaverse-based learning platforms have evolved higher education and have transformed the way higher education systems are taught and implemented. In healthcare education, these technologies could enhance students' comprehension of complicated knowledge by using realistic simulation strategies and three-dimensional visualization. This study was conducted to evaluate the technology readiness, students' prior experience and learning expectations regarding the deployment of the EON XR Metaverse platform in a healthcare service quality course. With a structured questionnaire, a quantitative cross-sectional pre-survey was conducted among 77 undergraduate students. It assessed digital readiness, availability of technology resources, exposure to immersive technologies and expectations about Metaverse-assisted learning. A descriptive statistical analysis indicates a high level that students have confidence and comfort in using digital technologies, which is a sign of very strong technological readiness. Moreover, participants were not familiar with AR/VR applications or the EON XR platform, indicating that immersive education was another area of study for most participants. Despite this lack of exposure, students expressed very favorable expectations towards Metaverse-based learning. They felt that EON XR would represent a more stimulating academic learning experience than traditional lectures, enhance visualization of healthcare service quality concepts and lead to better understanding. Overall, the findings revealed that students were willing and ready to work with Metaverse-enhanced education. Nonetheless, ensuring effective implementation will depend on proper orientation and technical assistance along with addressing differences in device accessibility and internet connectivity, to enhance both the learning process and equity in learning opportunities.

Keywords: *Metaverse, EON XR, Healthcare Service Quality, Digital Readiness, Student Expectations*

1. Introduction and Background

Education, however, has shifted drastically from the traditional, lecture-based model for academic education to immersive, technology-based educational frameworks. The "Metaverse," with special reference to its impact on the deployment of education in the education system, is a notion heavily developed after the COVID-19 pandemic, a worldwide disruption that led to a new way of teaching and learning that requires us to critically rethink, and then revolutionise, teaching and learning strategies globally (Lewis et al., 2024). This new digital horizon appears to be a 3D digital realm that leverages Virtual Reality (VR), Augmented Reality (AR), the internet and internet-related developments to form immersive, networked online experiences via the development of high-tech computing solutions (Popov et al., 2024; Chen, 2026). The Metaverse is the subject of high national ambition in countries like Malaysia, where education policies expect many user-friendly sites to be widely used in institutions. In this rapidly evolving and advanced ecosystem, platforms such as EON XR are adopting augmented and virtual reality to create experiential, hands-on learning environments connecting the physical and virtual worlds (Nguyen & Kim, 2024). For complex and applied fields namely, Healthcare Service Quality, spatial visualization and immersive scenario-based learning is a useful tool for bridging the gap between theory and practice (Bansal et al., 2022). Training in healthcare service quality is dependent upon students' understanding of complex organisational systems, patient-practitioner relationships and standardized clinical methods.

Simulation has become the backbone of healthcare education and training for generations as a potent and resource-demanding tool that reshapes medical education (Popov et al., 2024). Virtual reality and Metaverse technologies provide them with immersive learning environments where students can conduct ultra-realistic

simulations of care settings without having to jeopardize patient safety or the prohibitive capital spending required for the physical space (Lewis et al., 2024). The real ability to adopt such sophisticated tools is dependent heavily on end-user technological readiness and their infrastructure levels, and needs. However, there is a known readiness gap despite the potential benefits that increasing student engagement and cross-border collaboration could offer; this has largely depended on the relatively limited prior experiential exposure to immersive AR and VR technologies by students (Puiu & Udriștioiu, 2024). To fully integrate EON XR and other vast platforms into a healthcare service quality curriculum, we need to have a thorough understanding of the student's behavioural intent and technological self-efficacy. Model and assess factors such as user comfort, platform ease of use, compatibility with devices and interest of the learners to ensure a smooth user transition (Puiu & Udriștioiu, 2024).

The Healthcare Service Quality (HSM606) course is a key unit for the Semester 5 students attending the Bachelor of Health Administration course at Universiti Teknologi MARA (UiTM) Malaysia. The course is taught using an online course system on the internet through Google Meet, and in Semester 5, students are studying according to a flexible learning mode. The above-mentioned factors make online learning more flexible and accessible for both lecturers and students. The presence of online teaching environments has been criticised for providing low student participation and interaction, low engagement, and difficulty in sustaining their attention in the virtual learning environment (Martin & Bolliger, 2018; Bond et al., 2020; Dhawan, 2020). A particularly challenging aspect has to do with the course itself, i.e., the Healthcare Service Quality course. The syllabus provides access to extensive theory/health care quality frameworks, accreditation systems, patient-safety principles, and continuous quality improvement models requiring a significant amount of read/analysis and critical thinking. As a result, the student demographic often sees the subject as theoretical or a "dry course" where they receive little motivation to learn and often lack the ability to tie abstract theory to actual healthcare practice (Biggs et al., 2011; Means et al., 2014). Traditional courses for online learners may be thus inadequate for deep learning and learning to be learned in a meaningful way. To alleviate these challenges, the implementation of the EON XR Metaverse platform is an innovative strategy to integrate EON into teaching that integrates new teaching concepts based on experiential and constructivist learning theory. Emerging technologies, which provide rich and immersive educational experiences, allow students to see healthcare environments, interact in three-dimensional healthcare facilities, simulate real conditions in the healthcare environment and simulate quality management situations, that traditional online classes may be unable to present (Radianti et al., 2020), including Extended Reality (XR), Virtual Reality (VR), and Metaverse environments. These immersive educational experiences are conducive to active learning, collaboration and interactive experiences that result in enhanced student engagement in learning, which, in turn, contribute to enhanced students' comprehension, motivation and engagement (Makransky & Petersen, 2021).

Furthermore, the implementation of EON XR has the potential to connect theoretical learning with real-life encounters as students can encounter authentic healthcare service quality scenarios in a secure virtual environment. Given the burgeoning digitization of higher education, the use of the Metaverse for learning has the potential to improve teaching effectiveness and learning outcomes across higher education, especially in challenging classes, such as Healthcare Service Quality (Dwivedi et al., 2022; Tlili et al., 2022). Thus, the purpose of this study is to evaluate the technological readiness, experience and expectations of students when integrating the EON XR Metaverse platform into their coursework. Adopting a cross-sectional, pre-survey design covering three different academic cohorts, this study provides a comprehensive investigation into digital comfort and device accessibility, internet stability, and learning needs. The results will yield invaluable, data-driven insights on user behaviour to inform targeted teaching strategies to guarantee that the shift to Metaverse-based learning is equitable and meaningful.

2. Literature Review

The Evolution of Immersive Technologies in Higher Education

The Metaverse has radically transformed higher education into new learning environments that provide immersive, interactive, and dynamic classrooms as opposed to the usual classroom-based education context (Popov et al., 2024). This nascent digital realm encompasses VR, AR, and other immersive technologies to support virtual campuses, interactive simulations, and experiential learning scenarios (Lewis et al., 2024). The Metaverse is, in a sense, the next generation of the Internet, where users experience a high degree of spatial

presence embedded in a complex network of immersive virtual environments (Bansal et al., 2022). The growth of hardware and software to enable Metaverse development is speeding up, giving students access to a richer, more lifelike learning experience than with traditional virtual or augmented reality education.

The Metaverse concept has received a lot of buzz in the world of higher education in the past couple of years. It is a tool for the pooling of knowledge across disciplines through virtual learning environments where students from totally distinct academic backgrounds can collaborate, exchange ideas and collaborate on multidisciplinary projects (Lewis et al., 2024). The Metaverse brings students closer to actual practical situations in the real world, reducing physical restrictions, enabling an increased depth of learning through learning-by-doing approaches (Popov et al., 2024). Moreover, it presents unparalleled avenues for global collaboration as students and lecturers can interact across geographic boundaries, overcoming the geographical barriers of distance between campuses.

Virtual Reality and Simulation in Healthcare Education

Simulation is an efficient and resource-intensive technology that has revolutionized medical and healthcare education, transforming it altogether over the past decade (Popov et al., 2024). In medical education, it has been suggested that virtual reality (VR) is a new simulation technology that may deliver simulation in a more advanced manner that provides multiple advantages for medical educators and students by enabling affordable, standardized, repeatable, and on-demand technology in the medical arena (Bansal et al., 2022). VR is a computer-generated virtual world experienced through a variety of technical devices such as motion tracking, spatial audio, tactile hand controllers, and head-mounted displays (HMDs) that can produce 360-degree first-person views. These very modern products make the virtual environment incredibly immersive by providing a means for active participation, beyond the confines of traditional screens.

Past studies have repeatedly demonstrated the far-reaching potential of VR as a learning platform in medical and healthcare, explicitly discussing improved educational outcomes, increased speed of skills acquisition, greater retention, and the encouragement of hands-on learning (Lewis et al., 2024). VR has provided healthcare students the opportunity to experience realistic clinical and operational environments without placing patient safety in danger (Bansal et al., 2022). In a systematic review addressing teaching efficiency with a comparison between VR and traditional education, it was found that VR training led to a statistically significant improvement in students' clinical skills and practical performance (Popov et al., 2024). Research comparing HMD simulations with traditional teaching models (e.g., standard slides or linear videos) demonstrated that immersive VR dramatically accelerated the learning curve and heightened overall learner motivation.

The EON XR Platform: Capabilities and Pedagogical Impact

EON XR is a full-fledged platform that integrates exploration, creation, and guidance in fully immersive XR environments (Nguyen & Kim, 2024). A full-fledged Augmented and Virtual Reality application with features developed in-depth for remote learning and training in a high-level, hands-on and immersive environment. The program operates as a global, mobile-first ecosystem where academic skills are incorporated into real-world situations through immersive learning experiences and structured entrepreneurial routes. One of the key advantages of EON XR is its intuitive and completely code-free platform, which is designed to encourage both students and instructors to create content for deeper learning and engagement (Nguyen & Kim, 2024). By means of its intuitive and entirely code-free platform that fosters an intensively rich proprietary library of digital assets, users can rapidly make dynamic educational content in an instant on their mobile device, tablet, PC, or VR headset using a vast proprietary collection of digital content and resources.

Additionally, the platform is highly integrated with AI based on intelligent deep learning capabilities to improve the learning productivity of the educational processes. Its built-in spatial AI functionality is capable of auto-generating intricate lessons based on a user's course catalogue (enriched with vibrant images, targeted subtopics, rich media, and personalized quizzes). It also contains advanced AI avatars reflecting real-time personas for soft skills training, with intelligent feedback through complex simulations in context and real time. This level of interactivity is beyond what is possible today, as healthcare instructors can develop procedural training or assessment modules that automatically help train students through complex tasks in a relatively unscripted manner and contribute to the comprehension and acquisition of a skill and users can easily complete and acquire new knowledge (Bansal et al., 2022).

Technology Readiness and the Technology Acceptance Model (TAM)

Even though Metaverse technologies promise powerful theoretical advantages, their integration and implementation largely depend on users' involvement, readiness, and positive behavioural intentions (Puiu & Udriștioiu, 2024). It is for educational institutions to give attention, first and foremost, to addressing the technical and social, psychological, and motivational dimensions of the Metaverse before it's even operational. New quantitative work based on the Technology Acceptance Model (TAM) has extensively examined university students' readiness to support metaverse platforms, discovering that prominent analytical constructs are technological self-efficacy, perceived usefulness, perceived ease of use, general attitudes, and behavioural intentions (Puiu & Udriștioiu, 2024). For example, some studies demonstrate that students often demonstrate high levels of above-neutral technological self-efficacy and claim to find metaverse tools to be very helpful, but their hands-on experience with AR/VR hardware is still significantly low. This disparity marks a critical readiness gap in higher education for the adoption of immersive technology.

Developing positive attitudes and high digital confidence is a crucial prerequisite for successful metaverse use in education, with attitudes and perceived behavioural control being identified as major predictors of student behavioural intention to use the technology (Puiu & Udriștioiu, 2024). Comfort, convenience, compatibility, interest, and effectiveness — researchers strongly emphasize the concept of breaking down readiness to use technology into these distinct variables. Comfort is a precise measure of the user's reaction to, and ease of use while using the Metaverse in education and learning. Convenience specifically measures how convenient using simulations within a live classroom lesson to enact and engage with. The compatibility aspect closely monitors how the students' and the lecturer's reactions change when adapting to the virtual environments of these new environments. These are the critical variables, that it is possible to measure prior to deployment, which may help educators customize their onboarding and manage the technology adjustment.

Challenges, Barriers and Infrastructural Considerations

The Metaverse adoption in academia is undoubtedly not without its significant limitations. The primary hurdles faced by institutions mostly consist of technical, social, and ethical elements, as well as pedagogical difficulties largely associated with the fit with ongoing curricula and continued content creation (Lewis et al., 2024). The challenge with broad adoption is that infrastructural restrictions consistently pose a significant hurdle to widespread adoption, so the first step before any implementation is to assess user access to powerful hardware devices and extremely stable internet connections. Initiating such data collection, many users report that meaningful access is often hard to obtain without friction to access on devices with older hardware, rendering the platform unreachable for a small segment of the population. In addition, the effectiveness of VR training is dependent solely on balancing technical usability against the cognitive demands on the user (Popov et al., 2024). Repeatedly, participants in various VR-based studies, have mentioned usability barriers surrounding poor accessibility, a high frequency of technological failures, limited variability of training situations, and so on.

Cognitive load in virtual reality (VR) space correlates highly with divided attention, painful physical experiences like sickness in simulator setup, and overall need for better guidance and platform knowledge (Popov et al., 2024). In some VR formats, the disconnect between this digital world and the real-world movement of the user can cause a deep sense of nausea. Thus, enhancing immersive VR training in healthcare is inherently related to increasing the design variability and establishing structured post-simulation debriefing sessions to optimize learning outcomes and also significantly minimize cognitive overload. Moving to a Metaverse implementation within teaching and learning in a university requires extensive time for adjustment, extreme patience, and careful onboarding. The transition from the offline classroom to the 3D realm is jarring for users being used to traditional face-to-face learning, and using standard two-dimensional online meeting applications. It is therefore very important to grasp the sensitive juncture of the student level for technology readiness and state-of-the-art capabilities of modern platforms such as EON XR, along with the demanding factors of good quality education in healthcare. This study successfully fills the gap in academic publications by offering firsthand data on learners' expectations and readiness before formal Metaverse integration that is directly empirical.

3. Research Methodology

Research Design

Such an empirical descriptive, cross-sectional quantitative study was used here. This method was selected because it is most efficient for understanding attitude, technological readiness, and expectation at a single point in time before an educational intervention. The quantitative pre-survey approach permits a structured collection of numerical data about a defined student body which is critical to identify a basic comprehension of technological self-efficacy and infrastructural access prior to the integration of complex immersive technologies (e.g., EON XR Metaverse platform). The cross-sectional nature of the design means that all data related to the participants' readiness to utilize Augmented Reality (AR) and Virtual Reality (VR) in the learning of Healthcare Service Quality was collected shortly before the start of the intended curriculum modules. This allows for the genuine, preconceived notions and prior experiences of the students to be isolated without intervention effects.

Participants and Sampling Strategy

The population for this study included students in undergraduate studies who were currently taking a Healthcare Service Quality module at university. Purposive sampling was used to pick the people who belonged solely to selected academic groups that studied the Metaverse-enhanced learning curriculum. The sample consisted of 77 people from three different classes for a Bachelor's degree in Health Administration, UiTM Puncak Alam. This sample size fit for exploratory baseline analytics and descriptive statistics analysis. The inclusion criteria required that participants be active undergraduate students without severe visual or cognitive impairments that would completely prevent the use of screen-based or immersive technology.

Research Instrument and Variables

Data were collected utilizing a structured, self-administered digital questionnaire titled "Pre-Survey: Knowledge on Application of Metaverse via EON XR in Learning for Healthcare Service Quality." The survey instrument was specifically designed to evaluate the practical and psychological readiness of the students. The questionnaire was divided into distinct sections mapping to the study's core variables, primarily measured on a standard 5-point Likert scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree).

Section A: Demographic Information

This section contained the details of the respondents who participated in this study such as gender, student ID and name of class group.

Section B: Technological Readiness & Prior Experience

This core section consisted of five targeted items assessing the respondents' baseline digital competency and hardware capabilities:

General Digital Adoption: "I am comfortable adopting new digital technologies for educational purposes." (Evaluates general technological self-efficacy).

Immersive Technology Experience: "I have prior experience using Augmented Reality (AR) or Virtual Reality (VR) applications." (Establishes baseline familiarity with the medium).

Platform Familiarity: "I have previously used the EON XR platform or similar Metaverse learning environments." (Measures specific prior exposure to the intended educational software).

Device Accessibility: "I have access to a device (smartphone, tablet, or PC) capable of running AR/VR applications smoothly." (Assesses physical infrastructural readiness).

Internet Stability: "My current internet connection in Malaysia (campus or home) is stable enough for downloading and running 3D models/virtual environments." (Assesses network infrastructural readiness, a critical component for cloud-based spatial computing).

Section C: Expectations

This section featured two items designed to capture the students' perceived usefulness and pedagogical expectations of the Metaverse integration:

Anticipated Engagement: "I expect that learning through the Metaverse (EON XR) will be more engaging than traditional lectures."

Pedagogical Efficacy (Visualization): "I believe immersive AR/VR scenarios will help me better visualize and understand healthcare service quality concepts."

Data Collection Procedure

Data were gathered among registered students of the Healthcare Service Quality course at Universiti Teknologi MARA (UiTM), Malaysia. Data were collected over the first week of the lectures so that an accurate baseline of student readiness was established before any formal exposure to the Metaverse curriculum. The questionnaire was distributed online (Google Forms) to allow prompt accessibility and capture of the data. The survey link was distributed to the students through their official academic communication channels (Ufuture and WhatsApp).

The pre-survey participation was entirely voluntary, and students were properly briefed about the purpose of the data collection procedures. It was also noted that responses would be used to customize the onboarding of the technology and not positively or negatively influence their academic grades. Responses were automatically timestamped and securely logged into an encrypted spreadsheet (Pre-Survey_ Knowledge on Application of Metaverse via EON XR in Learning for Healthcare Service Quality (Responses).xlsxpre survey.xlsx) to maintain data integrity and respondent confidentiality.

Data Analysis

After the survey window was closed, the raw data was extracted and evaluated in Python with the explicit use of the pandas library for effective data manipulation and statistical computation. The cleaning procedure verified completion of responses, anomaly detection, and reorganization of column headers for programmatic computation. All 77 surveys contained no incomplete surveys or missing data points (NAs) in the core Likert-scale variables.

Descriptive statistics were used as the main statistical methodology. The analysis computed:

Frequencies and Percentages: To show the demographic spread among the three academic groups.

Measures of Central Tendency (Mean): To analyse the mean agreement of the student body for every variable of technological readiness and expectation.

Measures of Dispersion (Standard deviation): To measure the spread of the responses (volatility) because this is essential to diagnose possible distributional inequalities in the infrastructure (e.g., wide range of devices or unstable internet).

The outcome of these descriptive measures supplied a rich, quantitative baseline for the latter discussion and pedagogy recommendations pertaining to the implementation and delivery of the EON XR platform.

4. Results

Overview of the Respondents

A total of 77 valid responses were obtained from undergraduate students who completed the pre-survey before participating in the EON XR Metaverse learning activities. Regarding the gender of the respondents, the majority of them, 91% (70 students) were female while only 9% (7 students) were male. The results also indicated that 35.1% (27 students) of the respondents were recorded from Group BA2654A while 29.9% (23 students) were from BA2654B, and 35.1% (27 students) of them were from BA2654C.

Reliability Analysis

Table 1: Reliability Analysis of the Study Constructs (N = 77)

Construct	Number of Items	Cronbach's α	Reliability Interpretation
Technological Readiness & Prior Experience	5	0.86	Good
Expectations	2	0.92	Excellent

Note. Reliability interpretation follows commonly accepted thresholds: $\alpha \geq .90$ = Excellent; .80–.89 = Good; .70–.79 = Acceptable (Hair et al., 2022; Taber, 2018).

This questionnaire's internal consistency was assessed using Cronbach's alpha (α) to test the reliability of measurement scales. Based on Table 1, the Technological Readiness and Prior Experience construct, consisting of five items, achieved a Cronbach's alpha coefficient of 0.86, indicating good internal consistency. This

indicates that the items consistently reflect students' readiness to accept digital environments and their prior experiences with immersive technologies like AR, VR, and EON XR. The Expectations construct, having two items, exhibited a Cronbach's alpha of 0.92, meaning that the internal consistency was strong. The two items measuring whether students expected Metaverse-based learning to be effective or engaging exhibited a high reliability coefficient. The outcome indicates consistency in respondents' answers for the expectation items, confirming that the construct accurately reflects students' thoughts about learning through EON XR.

Overall, both constructs were above the cut-off threshold of 0.70, denoting acceptable reliability for use in practice (Hair et al., 2022). Findings also confirm that the measurement instrument has good internal consistency and is fit for follow-up statistical analyses such as descriptive statistics, correlation analysis, and regression analysis. In addition, the excellent reliability found for the Expectations construct enhances confidence in the robustness of the responses' reliability, whereas good reliability for the Technological Readiness and Prior Experience construct suggests that the items appropriately reflect a single construct. Thus, the questionnaire is seen as a relevant tool to assess undergraduate students' readiness and expectations for implementing the Metaverse (EON XR) technology into healthcare service quality education.

Descriptive Analysis

Table 2: Descriptive Statistics (N = 77)

Item	Mean	SD	Interpretation
Comfortable adopting new digital technologies for education.	4.35	.739	Very High
Prior experience using AR/VR applications	1.92	.791	Low
Previously used EON XR or similar Metaverse platforms	2.01	.716	Low
Have access to a suitable smartphone/tablet/PC	3.51	1.143	Moderate-High
Internet connection is stable enough for AR/VR learning	3.58	1.030	Moderate-High
Expect Metaverse learning to be more engaging than lectures	3.81	.828	High
Believe AR/VR will improve understanding of healthcare service quality	3.97	.827	High

There were a few important patterns that emerged from descriptive statistics. First, students were found to be extremely digitally ready ($M = 4.35$), showing that they were relatively well equipped with the ability to embrace new educational technologies. But exposure to immersive technologies was pretty minor in the past. The students had little prior experience with AR/VR applications ($M = 1.92$) and similar low familiarity with the EON XR platform ($M = 2.01$). By contrast, respondents reported moderate to high access to digital devices ($M = 3.51$) and acceptable internet connectivity ($M = 3.58$), indicating that the majority of students had the technological infrastructure in place for engaging in immersive learning exercises. More importantly, students had positive expectations regarding the educational potential of Metaverse learning. EON XR was expected to be more engaging than traditional lectures ($M = 3.81$) and participants believed immersive technologies would improve their understanding of healthcare service quality concepts ($M = 3.97$).

Summary of Mean and Standard Deviation

Table 3: Results of Construct-Level Descriptive Analysis (N = 77)

Construct	<i>M</i>	<i>SD</i>	Interpretation
Technological Readiness & Prior Experience	3.08	0.49	Moderate
Expectations	3.89	0.78	High

The descriptive statistics of the two study constructs, Technological Readiness and Prior Experience and Students' Expectations towards Metaverse (EON XR) learning, are summarised in Table 3. Overall, the results revealed that the students exhibited a modest degree of technological readiness and prior experience ($M = 3.08$, $SD = 0.49$), as the respondents were open to digital technologies in general whereas they had little prior experience with technologies such as Augmented Reality (AR), Virtual Reality (VR), and the EON XR platform, with immersive learning technologies. This result hints at the realization that the students have a decent level of technological proficiency, but sometimes learning/orientation and training may be necessary before fully engaging with Metaverse learning.

On the other hand, students had high expectations of Metaverse learning implementation ($M = 3.89$, $SD = 0.78$). The greater mean score suggests that respondents predicted that the EON XR platform would lead to better engagement and better comprehension of ideas about healthcare service quality. Although they had only limited experience with immersive technologies, students also had a positive perception in terms of the educational opportunity of Metaverse-based learning and could be a useful teaching method. This indicates a strong intention to implement innovative instructional practices and demonstrates optimism among students toward the adoption of immersive technologies within higher education. Because the standard deviations for both constructs were relatively low ($SD = 0.49$ and $SD = 0.78$), it shows that the respondents' perceptions were somewhat consistent throughout the sample. Overall, these results reveal that although students may need additional exposure to immersive technologies, as a result of their positive expectations and positive attitudes toward digital innovation, students are appropriately positioned for Metaverse-based learning. These findings support preliminary evidence to establish EON XR as a novel teaching tool for education of healthcare service quality.

Detailed Analysis for Each Construct

Table 4: Detailed Analysis of the Study Constructs (N = 77)

Item	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
Comfortable adopting new digital technologies for education.		1 (1.3%)	9 (11.7%)	29 (37.7%)	38 (49.4%)
Prior experience using AR/VR applications.	27 (35.1%)	29 (37.7%)	21 (27.3%)		
Previously used EON XR or similar Metaverse platforms.	19 (24.7%)	38 (49.4%)	20 (26%)		
Have access to a suitable smartphone/tablet/PC.	5 (6.5%)	9 (11.7%)	21 (27.3%)	26 (33.8%)	16 (20.8%)
Internet connection is stable enough for AR/VR learning.	1 (1.3%)	11 (14.3%)	24 (31.2%)	24 (31.2%)	17 (22.1%)
Expect Metaverse learning to be more engaging than lectures.		2 (2.6%)	29 (37.7%)	28 (36.4%)	18 (23.4%)
Believe AR/VR will improve understanding of healthcare service quality.		2 (2.6%)	21 (27.3%)	31 (40.3%)	23 (29.9%)

Students' Technological Readiness

Table 4 shows the willingness of respondents to use digital technologies before they adopted the EON XR Metaverse platform. Almost half of respondents (49.4%) strongly agreed, and 37.7% agreed they were comfortable with new digital technologies used for educational purposes. Only one respondent (1.3%) disagreed, showing overall very high digital confidence among participants. These results imply that most undergraduate healthcare students already have enough digital literacy to accommodate the use of immersive educational technologies. Digital readiness is seen as one of the greatest forces affecting technology acceptance and successful digital transformation in higher education (Dwivedi et al., 2022). In turn, students with digital tools tend to be more comfortable using technology and experimenting with new technology, in particular VR, and the Metaverse. These results are also consistent with the Technology Acceptance Model (TAM), which suggests users who have a higher technology self-efficacy will see the new technologies and tools they adopt as useful and easier to use and therefore act accordingly (Davis, 1989; Venkatesh et al., 2022).

Previous Experience with AR/VR Technologies

Despite high levels of digital confidence, students were inexperienced in immersive technologies. About 72.8% disagreed or strongly disagreed that they had previous experience with AR or VR applications, and only 27.3% had a neutral answer. None of the respondents indicated prior agreement with it. For the same reason, previous exposure to EON XR and similar Metaverse learning environments was extremely limited. Almost three-

quarters (74.1%) disagreed or strongly disagreed about their use of EON XR, whereas only 26.0% picked a neutral reaction. These results show that the respondents are first-time users of immersive educational technologies. Such a trait bolsters the study as students' perceptions after the implementation are less influenced by their previous experience or familiarity.

Recent research consistently identifies a lack of practical experience for students in the educational Metaverse, despite increased institutional interest in adopting Metaverse technologies, particularly in developing countries and healthcare education (Radianti et al., 2020). This means that the lack of prior exposure provides a significant opportunity for universities to gradually embed immersive technology into curricula over time, while providing sufficient orientation and technical support.

Accessibility of Learning Infrastructure

Overall, the respondents are perceived to have adequate access to technological infrastructure. Over half (54.6%) generally agreed and strongly agreed with the claim that they had devices that could run AR/VR programs, while only 18.2% disagreed. Roughly a quarter (27.3%) remained neutral. Likewise, 53.3% of respondents said that their internet connection was stable enough to download and use Virtual Learning environments. But around one-third of respondents, 31.2%, were neutral in this regard, and 15.6% said they lacked good internet coverage.

These results indicate that the technology infrastructure is generally a decent enough base for supporting Metaverse learning implementation, but the internet stability level continues to be a risk for a few users. Stable internet connectivity has been consistently found to be one of the main factors contributing to the effectiveness of immersive learning as rendering three-dimensional environments demands a higher bandwidth than traditional online learning platforms (Evans et al., 2023). As such, educational institutions that utilize Metaverse-based teaching should explore aspects such as hybrid learning approaches, optimized media-rich education, and technical support to reduce interruptions brought on by network impairments.

Students' Expectations toward Metaverse Learning

On the other hand, students expressed very positive expectations concerning the educational impact of EON XR. In terms of learning engagement, 59.8% either agreed or strongly agreed, (whereas 2.6% disagreed) that learning through the Metaverse would be more engaging than conventional lectures. On the other hand, 37.7% were neutral. In a similar vein, 70.2% agreed that they believe immersive AR/VR scenarios would enhance their comprehension of healthcare service quality concepts, and 2.6% disagreed. The overwhelmingly positive expectations indicate a strong degree of acceptance by the students long before the tech is introduced to them. According to expectancy-value theory, learners who expect benefits from the learning experience are motivated to participate actively in the learning experience. These results are consistent with recent evidence showing that the combination of immersive learning environments and the experiential learning of students, which enables students to visualize abstract healthcare processes in realistic virtual environments, increases learner engagement, enhance conceptual understanding (Kye et al., 2023; Tlili et al., 2022).

Discussion

The current study aimed to understand the readiness and expectations of undergraduate healthcare students at the beginning of the implementation of the EON XR Metaverse application within healthcare service quality education. Overall, students have relatively low exposure to immersive technologies but show significant digital readiness and positive beliefs about the educational value of Metaverse-based studies. These findings corroborate earlier research indicating that students are becoming more open to new educational technologies despite their lack of hands-on experience (Dwivedi et al., 2022; Tlili et al., 2022). An additional finding is the high level of confidence in the digital environment. Almost 90% of those surveyed reported being comfortable implementing new digital technologies for educational purposes. This finding highlights the traits of today's Generation Z learners who come from technology-rich contexts and tend to adopt digital innovations better than previous generations (Prensky, 2001; Dwivedi et al., 2022). Digital confidence has consistently been identified as a significant predictor of technology acceptance, as students with stronger digital self-efficacy find new technologies more useful and intuitively easy to navigate and use (Venkatesh et al., 2022).

As per the Technology Acceptance Model (TAM), perceived ease of use and perceived usefulness play an

important role in predicting users' behavioural intentions to adopt educational technology (Davis, 1989). However, respondents reported virtually no prior experience with EON XR and low exposure to AR/VR technologies despite reporting high levels of digital confidence. Over 70% had not used AR/VR applications; close to three-quarters were novices to EON XR or similar Metaverse learning environments. This finding agrees with recent systematic reviews revealing that even though research into immersive learning is very much growing, the actual implementation in higher education has been quite restricted, especially in developing countries and healthcare education (Radianti et al., 2020). Similar results were reported by Tlili et al. (2022), who concluded that the educational Metaverse is still in its early adoption stage globally. The sparse experience seen in students offers opportunities for educators because post-implementation learners' perceptions are less likely to be influenced by familiarity and technological bias beforehand. Instead, learners will be evaluated primarily by their own authentic experiences of learning the platform.

Prior studies indicate that this first experience with immersive learning environments usually produces considerable levels of curiosity and motivation with educational design and technical support being sufficient (Kye et al., 2023). The results also suggest that subjects have a good level of tech infrastructure to support in-depth learning. Over half of the study participants claimed they had the appropriate devices to run AR/VR applications and a similar proportion said they had stable internet to be able to use virtual learning environments. However, around a third of them were unsure of the quality of the internet. This finding aligns with prior literature which reports the use of tech infrastructure and internet stability as significant obstacles to meaningful Metaverse implementation (Evans et al., 2023; Dwivedi et al., 2022). Higher education institutions can have accessible technical, online, and device support to integrate immersive learning. A surprising and perhaps the most positive finding may arise from the expectation of students about the educational benefits of EON XR. Over 60% thought learning through the Metaverse would be more interesting than a typical lecture, and more than 70% thought immersive AR/VR experiences would be useful to help them comprehend concepts of service quality in healthcare. This aligns with the constructivist framework for learning that the idea that learning is achieved when learners actively construct knowledge, rather than by passively receiving information (Popov et al., 2024).

Immersive technologies allow students an opportunity to learn through experiencing real-world virtual healthcare environments, engaging in simulated interactions, and visualizing abstract concepts that are difficult to explain using traditional methods of teaching (Kye et al., 2023). Healthcare service quality is integrally interconnected involving: communication, delivery of care, individual patient engagement, quality improvement, organisational processes and the delivery of the service. Such more complex interactions might not be well captured in traditional classrooms. The Metaverse offers experiential simulations, within which students learn about patient journeys, examine healthcare processes and practise decision-making in a realistic, virtualized environment - without compromising patient safety (Radianti et al., 2020). Such immersive experiences can enhance conceptual understanding, knowledge retention, learner engagement, and critical thinking skills (Tlili et al., 2022). At least from the perspective of the Technology Acceptance Model (TAM) students pre-conceptualize the value of Metaverse learning. Perceived usefulness is considered one of the best predictors of behavioural intention to adopt new educational technology in education (Davis, 1989; Venkatesh et al., 2022). As a result, these positive expectations contribute greatly to establishing EON XR in the implementation of EON XR in healthcare teaching.

The results of the current study also have implications with regard to the practical side for curriculum development. Universities that will undertake immersive learning must realise that digital literacy does not mean competence in immersive learning. Orientation, technical training, scaffolded learning activities, structured interventions and an ongoing stream of technical support all remain key to enhancing maximum learning experiences for students (Dwivedi et al., 2022; Tlili et al., 2022). Also, there should be continuous professional development in the form of faculty preparation for lecturers on how to develop effective pedagogical strategies for integrating immersive technologies with the traditional way of teaching to effectively use the technologies. Thus, the investigation shows promising baseline conditions for the implementation of EON XR in undergraduate healthcare education. Students have a high level of digital readiness, are sufficiently equipped with technological infrastructure and hold positive expectations even when they have little experience with immersive technologies. Overall, these results provide an auspicious basis for future investigations to examine the learning effectiveness of Metaverse technology on students' knowledge

acquisition, student engagement, critical thinking or healthcare service quality capability after actual implementation in reality (Kye et al., 2023; Radianti et al., 2020).

5. Implications and Conclusion

Educational Implications

This results in many implications for healthcare management education. Firstly, universities should not assume that students possess prior experience with immersive technologies simply because they are digitally competent. Thus, onboarding should be done in structured orientation sessions before immersive learning activities are performed. Second, curriculum developers must realize that students demonstrate strong enthusiasm toward innovative learning methods. This favourable stance is a crucial point from which to incorporate Metaverse technologies into education in high-quality healthcare service settings. Third, institutions will need to continue investing in their digital infrastructure and training their lecturers. The effectiveness of such a concept is contingent on more than simply the availability of technology; educators require the ability to design meaningful immersive learning experiences that reflect course learning outcomes.

Conclusion

This study provides important baseline evidence regarding undergraduate students' technological readiness and expectations toward the integration of the EON XR Metaverse platform in Healthcare Service Quality education. The findings indicate that students demonstrated a high level of digital confidence, with strong comfort in adopting new digital technologies for educational purposes. However, this readiness was not accompanied by substantial prior exposure to AR/VR or EON XR, highlighting a clear distinction between general digital competence and immersive technology experience. This finding is consistent with recent research suggesting that although immersive technologies are receiving increasing attention in higher education, their practical adoption remains at an early stage, particularly in healthcare education (Radianti et al., 2020; Tlili et al., 2022). Despite limited prior experience, students expressed high expectations toward Metaverse-based learning, particularly in relation to engagement and the visualization of complex healthcare service quality concepts. The findings therefore support the potential of immersive technologies to facilitate experiential and constructivist learning by enabling students to interact with realistic virtual environments and connect theoretical concepts with practical situations (Kye et al., 2023; Tlili et al., 2022). From the perspective of the Technology Acceptance Model, students' positive perceptions of the potential usefulness and accessibility of the technology may provide a favourable foundation for subsequent adoption (Davis, 1989; Puiu & Udriştoiu, 2024).

Nevertheless, successful implementation should not be assumed solely from students' digital confidence. The moderate levels of device accessibility and internet stability indicate that infrastructural inequalities may influence students' ability to participate effectively. Therefore, structured orientation, technical support, appropriate infrastructure, and lecturer preparedness are essential to ensure an inclusive and meaningful immersive learning experience (Dwivedi et al., 2022; Popov et al., 2024). Overall, the study establishes a positive baseline for implementing EON XR in healthcare service quality education while highlighting the need for appropriate pedagogical and technical preparation. Future research should compare pre- and post-intervention outcomes to determine whether positive expectations translate into measurable improvements in learning effectiveness, engagement, confidence, and satisfaction. Such evidence would further clarify the educational value of Metaverse-based learning in healthcare management education.

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